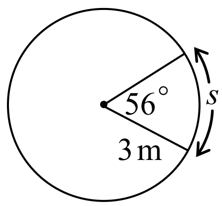
Question 1 T1

Use the information in the diagram to determine the value of the arc length "s", given the central angle of 56°.



Solution

$\theta = 56^\circ \times \dfrac{\pi}{180^\circ}$ 1 mark for conversion

$\theta = \dfrac{56\pi}{180}$ or $\dfrac{14\pi}{45}$

$s = \theta r$

$s = \left(\dfrac{14\pi}{45}\right)(3\,\text{m})$ 1 mark for substitution

$s = \dfrac{14\pi}{15}\,\text{m}$ or $2.932\,\text{m}$

2 marks

Question 2 T5

Solve $\tan^2\theta - 5\tan\theta + 4 = 0$ where $\theta \in \mathbb{R}$.

Solution

Method 1

$(\tan\theta - 1)(\tan\theta - 4) = 0$

$\tan\theta = 1 \qquad \tan\theta = 4$

$\qquad\qquad\qquad \theta_r = 1.3258$

1 mark for solving for $\tan\theta$ (½ mark for each branch)

$\theta = \frac{\pi}{4}, \frac{5\pi}{4} \qquad \theta = 1.326,\ 4.467$

$\theta = 0.785,\ 3.927$

2 marks (½ mark for each value of θ)

$$\left.\begin{aligned} \theta &= \frac{\pi}{4} + 2k\pi \\ &= 1.326 + 2k\pi \\ &= \frac{5\pi}{4} + 2k\pi \\ &= 4.467 + 2k\pi \end{aligned}\right\} \text{where } k \in \mathrm{I}$$

1 mark for general solution

4 marks

Method 2

$(\tan\theta - 1)(\tan\theta - 4) = 0$

$\tan\theta = 1 \qquad \tan\theta = 4$

$\qquad\qquad\qquad \theta_r = 1.3258$

1 mark for solving for $\tan\theta$ (½ mark for each branch)

$\theta = \frac{\pi}{4} \qquad \theta = 1.326$

$\theta = 0.785$

2 marks (1 mark for each value of θ)

$$\left.\begin{aligned} \theta &= \frac{\pi}{4} + k\pi \\ &= 1.326 + k\pi \end{aligned}\right\} \text{where } k \in \mathrm{I}$$

1 mark for general solution

4 marks

Question 3 R8, R10

Solve:

$$2^{5x} = 3(5)^{x-3}$$

Solution

$\log 2^{5x} = \log\left[3(5)^{x-3}\right]$	½ mark for applying logarithms
$5x\log 2 = \log 3 + (x-3)\log 5$	1 mark for product rule 1 mark for power rule
$5x\log 2 = \log 3 + x\log 5 - 3\log 5$	
$5x\log 2 - x\log 5 = \log 3 - 3\log 5$	½ mark for collecting like terms
$x(5\log 2 - \log 5) = \log 3 - 3\log 5$	
$x = \dfrac{\log 3 - 3\log 5}{5\log 2 - \log 5}$	½ mark for isolating x
$x = -2.009$	½ mark for evaluating a quotient of logarithms

4 marks

Question 4 P3

David and Sarah are in a class of 10 boys and 8 girls.
A committee of 3 boys and 2 girls is to be selected from the students in this class.

Determine the number of possible committees if David and Sarah cannot be on the same committee.

Solution

Method 1

All:	${}_{10}C_3 \times {}_8C_2 = 3360$	1 mark for all possible committees
Both:	${}_9C_2 \times {}_7C_1 = 252$	1 mark for both on the committee
	$3360 - 252 = 3108$	1 mark for subtraction of cases

3 marks

Method 2

Case 1: David, not Sarah	${}_9C_2 \times {}_7C_2 = 756$	½ mark for Case 1
Case 2: Sarah, not David	${}_9C_3 \times {}_7C_1 = 588$	½ mark for Case 2
Case 3: Neither David nor Sarah	${}_9C_3 \times {}_7C_2 = 1764$	1 mark for Case 3
	$756 + 588 + 1764 = 3108$	1 mark for addition of cases

3 marks

In the binomial expansion of $\left(\frac{3}{x^2} - x^5\right)^{10}$, simplify the 7th term.

Solution

$$t_7 = {}_{10}C_6\left(\frac{3}{x^2}\right)^4\left(-x^5\right)^6$$

$$t_7 = 210\left(\frac{81}{x^8}\right)\left(x^{30}\right)$$

$$t_7 = 17\ 010x^{22}$$

2 marks (1 mark for ${}_{10}C_6$, ½ mark for each consistent factor)

1 mark for simplification (½ mark for coefficient, ½ mark for exponent)

3 marks

Question 6 — R10

A lake affected by acid rain has a pH of 4.4.
A person suffering from heartburn has a stomach acid pH of 1.2.
The pH of a solution is defined as $\text{pH} = -\log\left[\text{H}^+\right]$ where $\left[\text{H}^+\right]$ is the hydrogen ion concentration.

How many times greater is the hydrogen ion concentration of the stomach than that of the lake?

Express your answer as a whole number.

Solution

Method 1

$$\text{pH} = -\log\left[\text{H}^+\right]$$

$$-\text{pH} = \log\left[\text{H}^+\right]$$

$$\left[\text{H}^+\right] = 10^{-\text{pH}}$$

1 mark for exponential form

$$\frac{\left[\text{H}^+\right]_{\text{stomach}}}{\left[\text{H}^+\right]_{\text{lake}}} = \frac{10^{-1.2}}{10^{-4.4}}$$

1 mark for comparison

$$= 10^{3.2}$$
$$= 1584.9$$
$$= 1585$$

2 marks

Method 2

Lake

$$4.4 = -\log\left[\text{H}^+\right]$$

$$-4.4 = \log\left[\text{H}^+\right]$$

$$10^{-4.4} = \left[\text{H}^+\right]$$

½ mark for exponential form

Stomach

$$1.2 = -\log\left[\text{H}^+\right]$$

$$-1.2 = \log\left[\text{H}^+\right]$$

$$10^{-1.2} = \left[\text{H}^+\right]$$

½ mark for exponential form

$$\frac{\left[\text{H}^+\right]_{\text{stomach}}}{\left[\text{H}^+\right]_{\text{lake}}} = \frac{10^{-1.2}}{10^{-4.4}}$$

1 mark for comparison

$$= 10^{3.2}$$
$$= 1585$$

2 marks

Question 7 T5

Solve the following equation algebraically over the interval $[0, 2\pi]$.

$$\cos 2\theta - 3\sin\theta - 2 = 0$$

Solution

$$\left(1 - 2\sin^2\theta\right) - 3\sin\theta - 2 = 0$$

1 mark for correct substitution of an appropriate identity

$$-2\sin^2\theta - 3\sin\theta - 1 = 0$$

$$2\sin^2\theta + 3\sin\theta + 1 = 0$$

$$(2\sin\theta + 1)(\sin\theta + 1) = 0$$

$$\sin\theta = -\frac{1}{2} \qquad \sin\theta = -1$$

1 mark for solving for $\sin\theta$ (½ mark for each branch)

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6} \qquad \theta = \frac{3\pi}{2}$$

2 marks for solutions (1 mark for each branch; ½ mark for each value in the left branch)

4 marks

Question 8 R12

Explain how the value of n affects the behaviour of the graph of the polynomial function $p(x) = (x+3)(x-1)^n$, as $p(x)$ approaches the x-intercept at $x = 1$.

Solution

If n is even, the graph will turn at the x-axis at $x = 1$.
If n is odd, the graph will cross the x-axis at $x = 1$.

1 mark

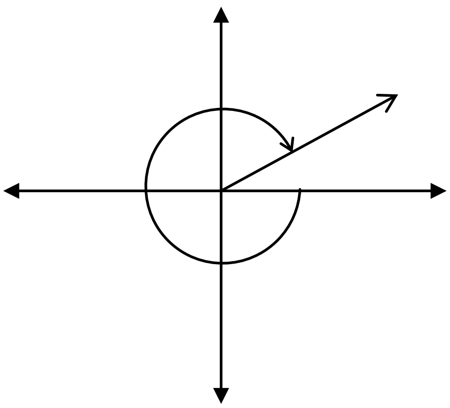
or

As n increases, $p(x)$ will become flatter around the intercept at $x = 1$.

Question 9 T1

Sketch the angle -320° in standard position.

Solution



1 mark for angle drawn in quadrant I

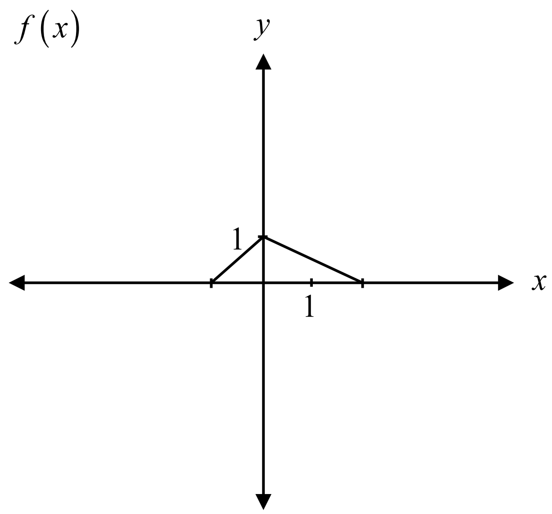
1 mark

Note(s):

- If directional arrow is not indicated, it is a communication error, E1 (final answer not stated).

Question 10 R5

Given the graph of $y = f(x)$, explain how to graph $y = f(-x)$.



Solution

Multiply the x-coordinates by -1.

1 mark

or

Reflect the graph over the y-axis.

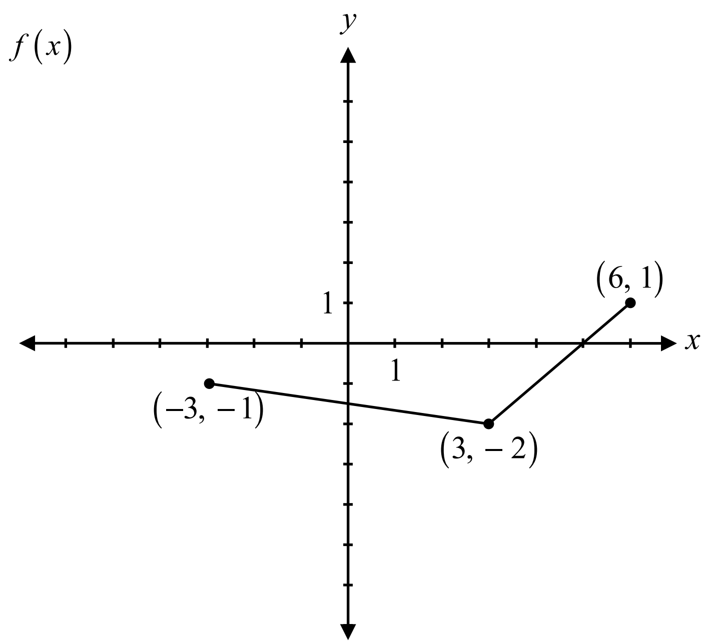
Question 11 R14

Explain how the graph of $y = \dfrac{3(x-1)}{(x-1)}$ is different than the graph of $y = 3$.

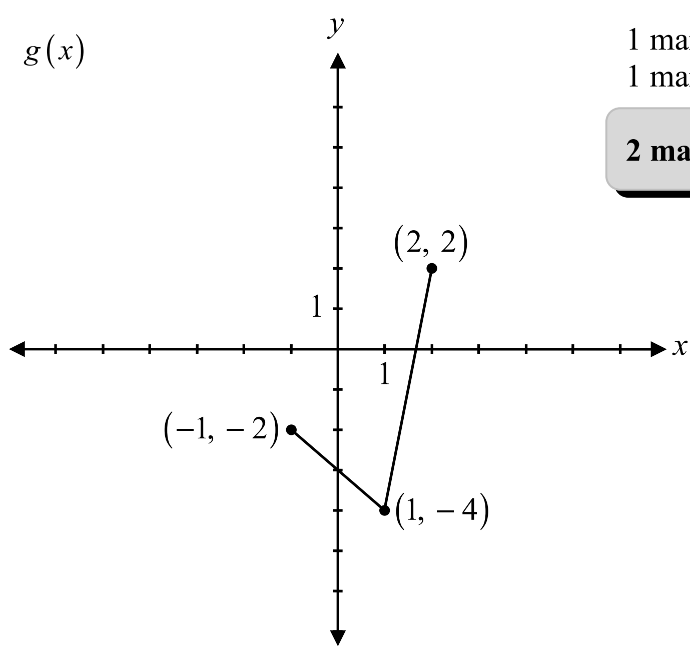
Solution

There is a point of discontinuity (hole) when $x = 1$ on the graph of $y = \dfrac{3(x-1)}{(x-1)}$. **1 mark**

Given the graph of $f(x)$, sketch the graph of $g(x) = 2f(3x)$.



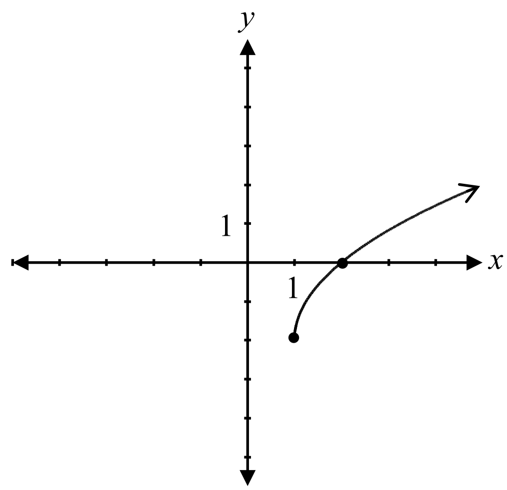
Solution



1 mark for horizontal stretch/compression
1 mark for vertical stretch

2 marks

Determine the equation of the radical function represented by the graph.



Solution

$y = 2\sqrt{x-1} - 2$

1 mark for vertical stretch
1 mark for horizontal translation
1 mark for vertical translation

3 marks

or

$y = \sqrt{4(x-1)} - 2$

1 mark for horizontal compression
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Question 14 P1

There are 2 types of pencils, 3 colours of highlighters, and 5 styles of pens.
If you must select one of each to form a set, how many different sets of writing instruments are possible?

a) 10

b) 11

c) 25

d) 30

Question 15 T2

The point $P(\theta)$ lies on the unit circle. What are the coordinates of the point P if $\theta = 120^\circ$?

a) $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

b) $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

c) $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

d) $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

Question 16 R13

Identify the function that has a domain of $\{x \mid x \geq 7\}$ and a range of $\{y \mid y \geq 0\}$.

a) $f(x) = \sqrt{x} + 7$

b) $f(x) = \sqrt{x} - 7$

c) $f(x) = \sqrt{x + 7}$

d) $f(x) = \sqrt{x - 7}$

Question 17 T4

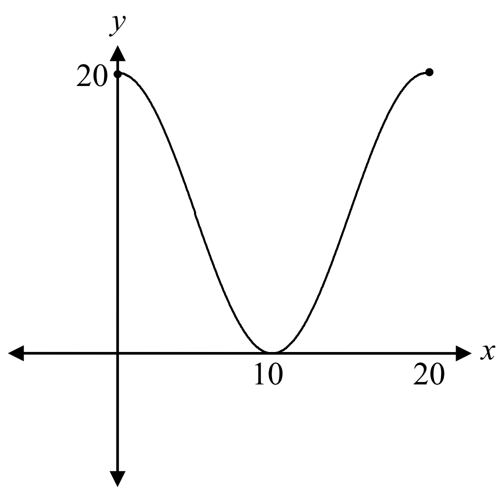
Using $y = -10\cos\left[\mathrm{B}(x - \mathrm{C})\right] + \mathrm{D}$, the value of C that corresponds to the following graph is:

a) 5

b) 10

c) 15

d) 20



Question 18 P4

Given the following row of Pascal's Triangle, identify the binomial expansion with these coefficients.

$$1 \qquad 5 \qquad 10 \qquad 10 \qquad 5 \qquad 1$$

a) $(x+y)^4$

b) $(x+y)^5$

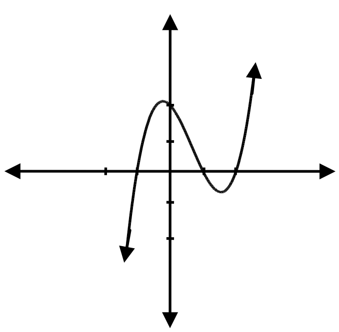
c) $(x+y)^6$

d) $(x+y)^7$

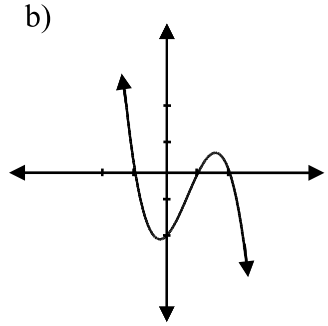
Question 19 R12

Identify the graph of the function $f(x) = -(x-2)(x-1)^2(x+1)$.

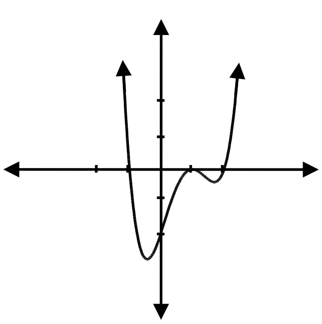
a)



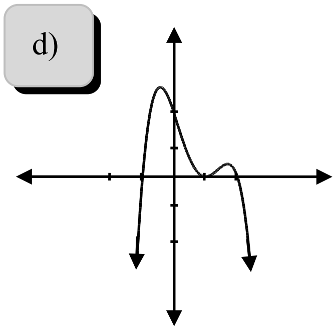
b)



c)



d)



Question 20 R7

Determine the value of $\log_9\left(\log_3 27\right)$.

a) $\frac{1}{3}$

b) $\frac{1}{2}$

c) 2

d) 3

Question 21 R4

If $(x,\ y)$ is a point on the graph of $y = f(x)$, identify the coordinates of this point on the graph of $g(x) = f(2x) + 5$.

a) $\left(\frac{x}{2},\ y+5\right)$

b) $(2x,\ y+5)$

c) $\left(\frac{x}{2},\ y-5\right)$

d) $\left(\frac{x}{2}-5,\ y\right)$

Question 22 R7, R8

Identify an equivalent expression for $1+\log_2 5$

a) $\log_2 5$

b) $\log_2 7$

c) $\log_2 10$

d) $\log_2 11$

Solve:

$$2\log_4 x - \log_4(x+3) = 1$$

Solution

$$2\log_4 x - \log_4(x+3) = 1$$

$$\log_4\left(\frac{x^2}{x+3}\right) = 1$$

1 mark for power rule
1 mark for quotient rule

$$4^1 = \frac{x^2}{x+3}$$

1 mark for exponential form

$$4x + 12 = x^2$$

$$x^2 - 4x - 12 = 0$$

$$(x-6)(x+2) = 0$$

$$x = 6 \quad \cancel{x = -2}$$

½ mark for solving for x
½ mark for rejecting extraneous root

4 marks

Question 24 R2, R5

The following transformations are applied to $f(x)$, resulting in a new function, $g(x)$.

- reflection over the y-axis
- horizontal translation of 3 units to the right
- vertical translation of 4 units down

Write the equation of $g(x)$ in terms of $f(x)$.

Solution

$g(x) = f\left(-(x-3)\right) - 4$

1 mark for horizontal shift
1 mark for vertical shift
1 mark for reflection

or

3 marks

$g(x) = f(-x+3) - 4$

Question 25 T4

The height of a bicycle pedal as the bicycle is moving at a constant speed can be represented by the following function:

$$h(t) = 15\cos\frac{2\pi}{5}t + 30$$

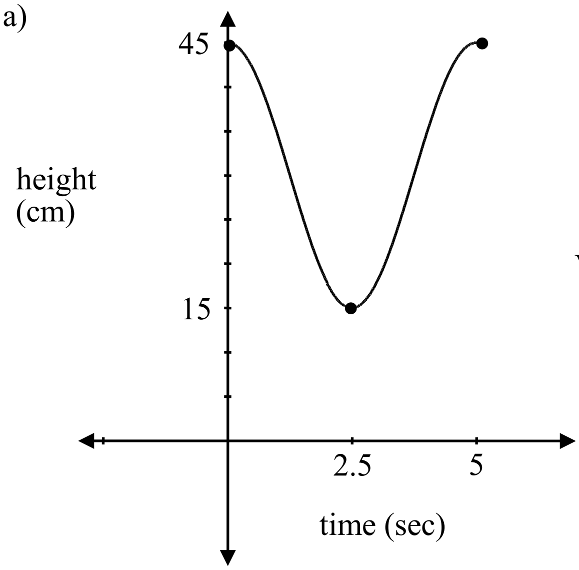
where h is the height of the pedal above the ground, in cm, and t is the time, in seconds.

a) Sketch a graph of at least one period of this function, where $t \geq 0$.

b) Determine the height of the bicycle pedal at 7.5 seconds.

Solution

a)



Amplitude $= 15$ — 1 mark for amplitude

$$\text{Period} = \frac{2\pi}{\frac{2\pi}{5}} = 5$$

1 mark for period

Vertical Shift $= 30$ — 1 mark for vertical shift

3 marks

b) From the graph:

$h(t) = 15$ cm

1 mark

or

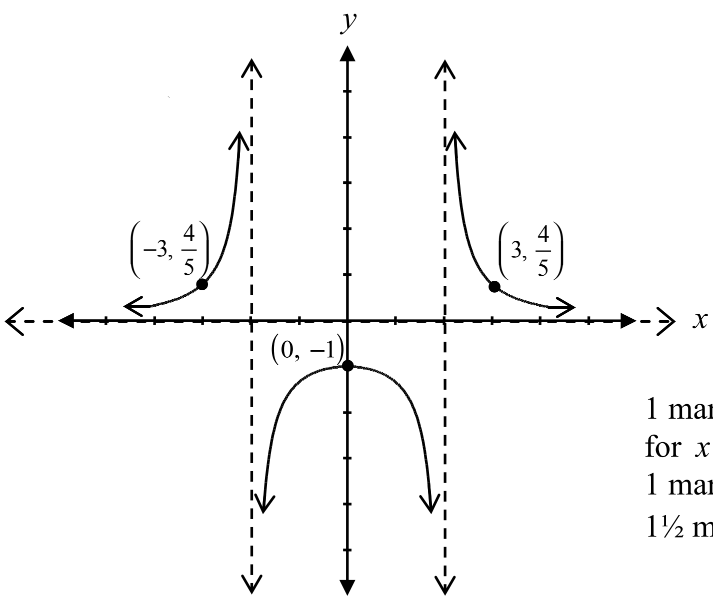
From the equation:

$$\begin{aligned} h(t) &= 15\cos\frac{2\pi}{5}(7.5) + 30 \\ &= 15\cos 3\pi + 30 \\ &= 15(-1) + 30 \\ &= 15 \text{ cm} \end{aligned}$$

Sketch the graph of the function $f(x)$ and determine the y-intercept.

$$f(x)=\frac{4}{(x-2)(x+2)}$$

Solution



1 mark for vertical asymptotes (½ mark for $x = 2$, ½ mark for $x = -2$)
1 mark for horizontal asymptote at $y = 0$
1½ marks for shape (½ mark for shape in each section)

y-intercept: -1

½ mark for y-intercept

4 marks

Question 27 R10

Kim solved the following logarithmic equation:

$$\log_2\left(-\frac{x}{3}\right) = \log_2(x-4)$$

$$-\frac{x}{3} = x - 4$$

$$-x = 3x - 12$$

$$-4x = -12$$

$$\cancel{x = 3}$$

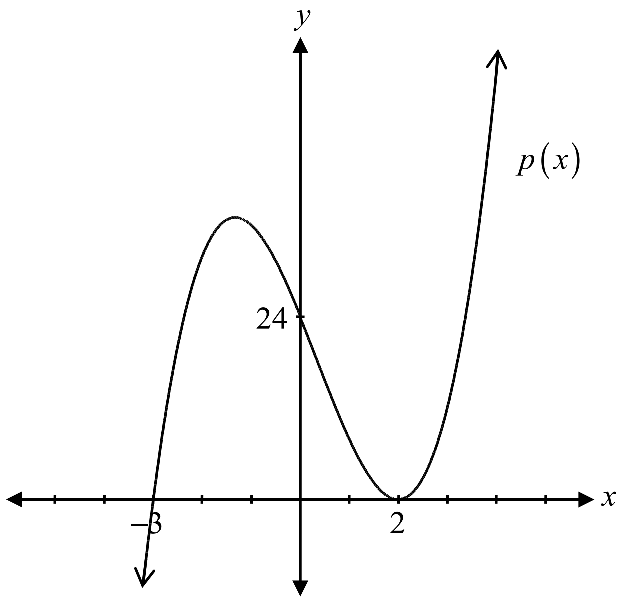
Explain why $x = 3$ is an extraneous solution.

Solution

$x = 3$ is an extraneous solution because the argument in a logarithmic equation cannot be negative.

1 mark

Determine the equation of the polynomial function represented by the graph.



Solution

$p(x) = a(x-2)^2(x+3)$

$24 = a(0-2)^2(0+3)$

$24 = a(4)(3)$

$24 = 12a$

$a = 2$

$p(x) = 2(x-2)^2(x+3)$

1 mark for x-intercepts (½ mark for each)
1 mark for multiplicity at $x = 2$

1 mark for leading coefficient

3 marks

Question 29 T1

Determine the coterminal angles with $\frac{2\pi}{3}$ over the interval $[-2\pi, 4\pi]$.

Solution

$-\frac{4\pi}{3}, \frac{8\pi}{3}$ 1 mark (½ mark for each coterminal angle)

1 mark

Question 30 R13

a) Solve the following equation:

$$0 = \sqrt{4x - 8} - 2$$

b) Explain how your answer in a) is related to the graph of $y = \sqrt{4x - 8} - 2$.

Solution

a)
$$4 = 4x - 8$$
$$12 = 4x$$
$$x = 3, \; x \geq 2$$

1 mark

b) The answer in a) is the x-intercept of the graph. **1 mark**

Note(s):

- $x \geq 2$ does not need to be shown.

Question 31 T3, T6

Determine the exact value of $\sin \frac{13\pi}{12}$.

Solution

$$\sin \frac{13\pi}{12} = \sin\left(\frac{3\pi}{4} + \frac{\pi}{3}\right)$$

1 mark for combination

$$= \left(\sin \frac{3\pi}{4}\right)\left(\cos \frac{\pi}{3}\right) + \left(\cos \frac{3\pi}{4}\right)\left(\sin \frac{\pi}{3}\right)$$

$$= \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right) + \left(-\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)$$

2 marks for exact values (½ mark for each)

3 marks

$$= \frac{\sqrt{2} - \sqrt{6}}{4}$$

Note(s):

- Other combinations are possible.

Question 32 R1

Given the functions $f(x) = x + 2$ and $g(x) = \dfrac{1}{x-5}$:

a) Determine the equation of the composite function $f(g(x))$ and its domain.

b) Determine the x-intercept and y-intercept of $f(g(x))$.

Solution

a) $$f(g(x)) = \frac{1}{x-5} + 2$$

or

$$f(g(x)) = \frac{2x-9}{x-5}$$

domain: $\{x \in \mathbb{R},\ x \neq 5\}$

1 mark for composition

1 mark for domain

2 marks

b) x-intercept: $\dfrac{9}{2}$

y-intercept: $\dfrac{9}{5}$

½ mark for x-intercept

½ mark for y-intercept

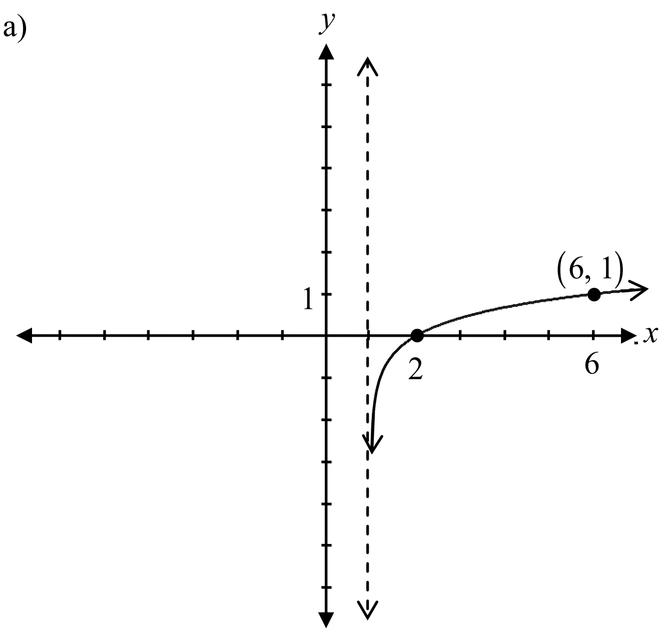
1 mark

a) Sketch the graph of $f(x) = \log_5(x-1)$.

b) Sketch the graph of $f^{-1}(x)$.

Solution

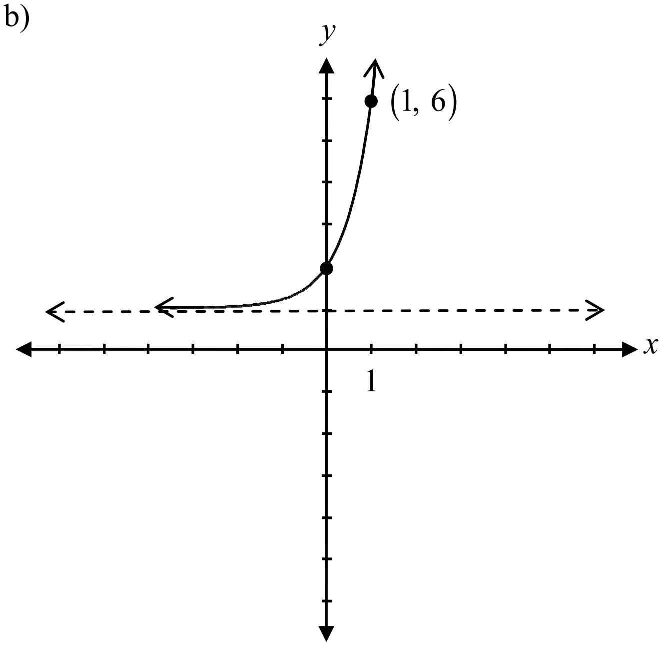
a)



½ mark for vertical asymptote at $x = 1$
½ mark for x-intercept at $x = 2$
½ mark for increasing logarithmic function
½ mark for consistent point on the logarithmic graph

2 marks

b)



1 mark for the graph of the inverse function consistent with a)

1 mark

Question 34 R14

Explain what the graph of a rational function looks like near a vertical asymptote.

Solution

The graph approaches infinity (positive or negative) as it approaches the asymptote.

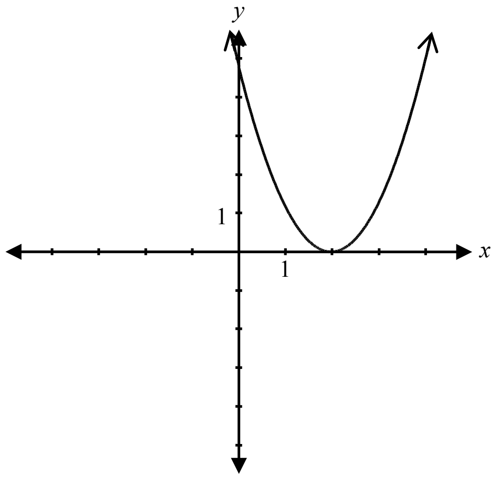
1 mark

or

The graph approaches the asymptote, but does not touch it.

Question 35 R6

Given the graph of $f(x) = (x-2)^2$,



determine one possible restriction for the domain of $f(x)$ so that its inverse is a function.

Solution

Domain: $[2, \infty)$

or

Domain: $(-\infty, 2)$

1 mark

Note(s):

- Other solutions are possible.

Question 36 T3,T6

Over the interval $[0,\ 2\pi]$, determine the non-permissible values of θ in the expression $\csc\theta(\cos\theta+1)$.

Solution

$\csc\theta = \dfrac{1}{\sin\theta}$	1 mark for correct substitution of an appropriate identity
$\sin\theta \neq 0$	½ mark for $\sin\theta \neq 0$
$\theta = 0,\ \pi,\ 2\pi$	½ mark for consistent non-permissible values

2 marks

Question 37 P3

Explain why ${}_3C_8$ is undefined.

Solution

In the formula ${}_nC_r$, the number of objects, n, must be larger than or equal to the number of objects selected, r.

1 mark

or

You can't select 8 objects from a total of 3.

Solve:

$$_{n}P_{3} = 48(n-1)$$

Solution

$$\frac{n!}{(n-3)!} = 48(n-1)$$

½ mark for substitution in correct formula

$$\frac{(n)(n-1)(n-2)\cancel{(n-3)!}}{\cancel{(n-3)!}} = 48(n-1)$$

1 mark for expansion of factorial
½ mark for simplification of factorials

$$n(n-2) = 48$$

$$n^2 - 2n - 48 = 0$$

$$(n-8)(n+6) = 0$$

$$n = 8 \quad \cancel{n = -6}$$

½ mark for solving for both values of n
½ mark for rejecting extraneous solution

3 marks

Christine dives off a diving board.
Her dive is modelled by the function $h(t) = t^3 - 3t^2 - t + 3$, where h is her height in metres, relative to the water surface and t is the time in seconds after diving off the diving board.

a) Given that $(t+1)$ is a factor for the function $h(t)$, determine the other factors.

b) Sketch the graph of the function $h(t)$ for the time interval $t = 0$ to $t = 3$.

c) Determine how many seconds Christine is underwater.

Solution

a)
$$\begin{array}{r|rrrr} -1 & 1 & -3 & -1 & 3 \\ & \downarrow & -1 & 4 & -3 \\ \hline & 1 & -4 & 3 & 0 \end{array}$$

½ mark for $t = -1$

1 mark for synthetic division (or equivalent strategy)

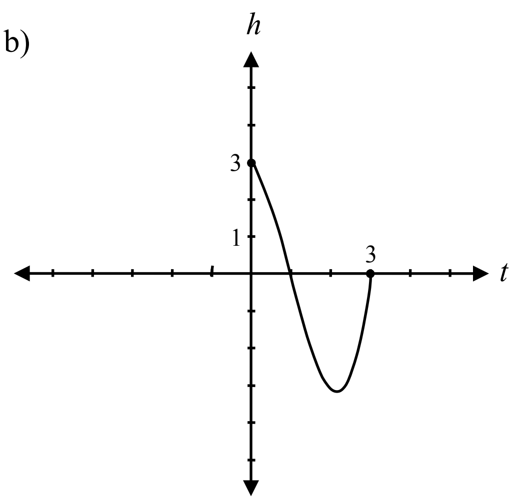
$h(t) = (t+1)(t^2 - 4t + 3)$

$h(t) = (t+1)(t-1)(t-3)$

½ mark for determining other factors

2 marks

b)



½ mark for t-intercepts
½ mark for h-intercept

1 mark

c) Christine is underwater for 2 seconds.

1 mark

Question 40 T6

Prove the identity for all permissible values of x.

$$\sec x + \tan x = \frac{\cos x}{1 - \sin x}$$

Solution

Method 1

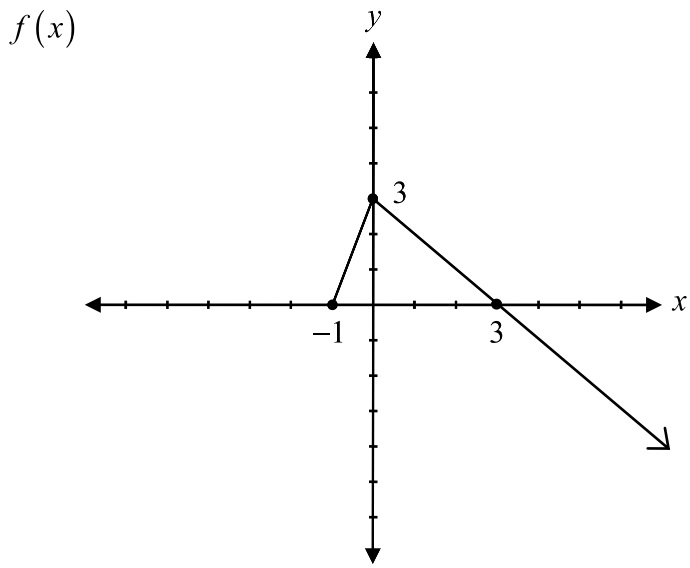
Left-Hand Side	Right-Hand Side
$\sec x + \tan x$	
$\dfrac{1}{\cos x} + \dfrac{\sin x}{\cos x}$	1 mark for correct substitution of appropriate identities
$\dfrac{(1+\sin x)(1-\sin x)}{\cos x(1-\sin x)}$	1 mark for logical process to prove the identity
$\dfrac{1-\sin^2 x}{(1-\sin x)\cos x}$	
$\dfrac{\cos^2 x}{(1-\sin x)\cos x}$	1 mark for algebraic strategies
	3 marks
$\dfrac{\cos x}{1-\sin x}$	

Method 2

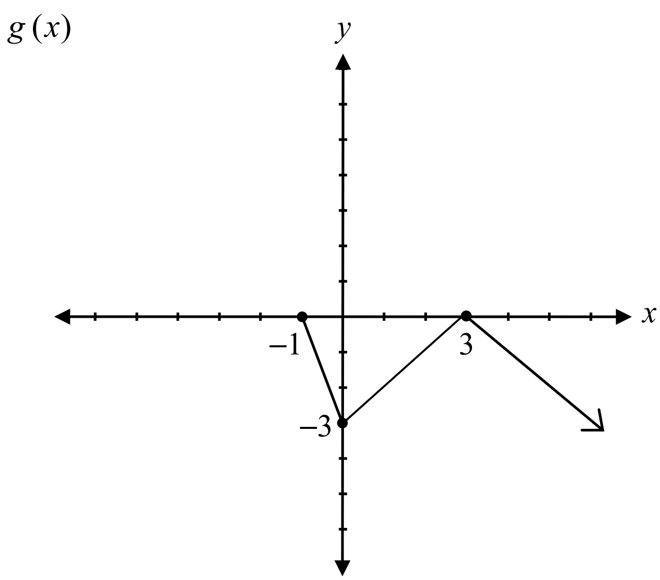
$$\sec x + \tan x = \frac{\cos x}{1 - \sin x}$$

Left-Hand Side	Right-Hand Side	
	$\frac{\cos x}{1 - \sin x}$	
	$\frac{\cos x(1 + \sin x)}{(1 - \sin x)(1 + \sin x)}$	1 mark for logical process to prove the identity
	$\frac{\cos x(1 + \sin x)}{1 - \sin^2 x}$	
	$\frac{\cos x(1 + \sin x)}{\cos^2 x}$	1 mark for correct substitution of appropriate identities
	$\frac{1 + \sin x}{\cos x}$	
	$\frac{1}{\cos x} + \frac{\sin x}{\cos x}$	1 mark for algebraic strategies
	$\sec x + \tan x$	**3 marks**

Given the graph of $f(x)$, sketch the graph of the function $g(x) = -|f(x)|$.



Solution



1 mark for absolute value
1 mark for vertical reflection

2 marks

Question 42 T2

Given that $\cot\theta = -\frac{2}{5}$, where θ is in Quadrant IV, determine the exact value of $\sin\theta$.

Solution

$P(2, -5)$	
$r^2 = (2)^2 + (-5)^2$	½ mark for substitution of $x = 2,\ y = -5$
$r^2 = 4 + 25$	½ mark for solving for r
$r = \sqrt{29}$	
$\sin\theta = \frac{-5}{\sqrt{29}}$	1 mark for $\sin\theta$ (½ mark for quadrant, ½ mark for value)

2 marks